

# What is the right geometry for minimization of MMD?

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## 1 Introduction

Given a functional  $\mathcal{F} : \mathcal{P}(\mathbb{R}^d) \rightarrow \mathbb{R}$ , its gradient-flow dynamics depend on the choice of geometry on the underlying space of measures. One of the most commonly used geometries is the Wasserstein-2 geometry, induced by the Wasserstein-2 distance. Its associated Onsager operator, which is the inverse of the Riemannian metric tensor, is given by  $\mathbb{K}_W(\rho) : T_\rho^* \mathcal{M}^+ \rightarrow T_\rho \mathcal{M}^+, \xi \mapsto -\operatorname{div}(\rho \nabla \xi)$ . Hence, the Wasserstein gradient flow of  $\mathcal{F}$  is

$$\partial_t \mu_t = \nabla \cdot (\mu_t \nabla \mathcal{F}'[\mu_t]). \quad (\text{W-GF})$$

Similarly, the Fisher–Rao geometry is induced by the Hellinger distance, with associated Onsager operator  $\mathbb{K}_{\text{FR}}(\rho) : T_\rho^* \mathcal{M}^+ \rightarrow T_\rho \mathcal{M}^+, \xi \mapsto \rho, \xi$ . Consequently, the Fisher–Rao gradient flow of  $\mathcal{F}$  is

$$\partial_t \mu_t = -\mu_t \left( \mathcal{F}'[\mu_t] - \int \mathcal{F}'[\mu_t] d\mu_t \right). \quad (\text{FR-GF})$$

An interpolation between these two dynamics is given by the well-known Wasserstein–Fisher–Rao gradient flow:

$$\partial_t \mu_t = \alpha \nabla \cdot (\mu_t \nabla \mathcal{F}'[\mu_t]) - \beta \mu_t \left( \mathcal{F}'[\mu_t] - \int \mathcal{F}'[\mu_t] d\mu_t \right). \quad (\text{WFR-GF})$$

Recently, a different geometry is considered in [Zhu \[2024\]](#), [Zhu and Mielke \[2024\]](#), [Gladin et al. \[2024\]](#), which uses the MMD as the geometry. The associated Onsager operator is  $\mathbb{K}_{\text{MMD}}(\rho) = \mathbb{K}_{\text{FR}}(\rho) \circ \mathcal{K}_\rho^{-1}$ . The Interaction-Force Transport (IFT) gradient flow is described as the following:

$$\partial_t \mu_t = -\mathbb{K}_{\text{MMD}}(\mu_t) \mathcal{F}'[\mu_t] = -\mu_t \mathcal{K}_{\mu_t}^{-1} \mathcal{F}'[\mu_t] = -\mathcal{K}^{-1} \mathcal{F}'[\mu_t]. \quad (\text{IFT-GF})$$

Similarly as WFR-GF, an interpolation of WGF and IFT-GF can also be constructed

$$\partial_t \mu_t = \alpha \nabla \cdot (\mu_t \nabla \mathcal{F}'[\mu_t]) - \beta \mathcal{K}^{-1} \mathcal{F}'[\mu_t]. \quad (\text{WIFT-GF})$$

## 2 $\mathcal{F} = \text{MMD}$ and geometry=MMD

We focus on MMD due to its interpretation as the worst-case integration error in the unit ball of a RKHS. If  $\mathcal{F}$  is the MMD distance  $\mathcal{F}(\cdot) = \text{MMD}^2(\cdot, \pi)$ , then  $\mathcal{F}'[\mu](\cdot) = \int k(x, \cdot) d(\mu - \pi) = \mathcal{K}(\mu - \pi)$ . MSIP uses the WFG-GF, but here we focus on the WIFT-GF. The main reason is that, [Gladin et al. \[2024\]](#) has proved that  $\text{MMD}^2(\mu_t, \pi) \leq e^{-2\beta t} \cdot \text{MMD}^2(\mu_0, \pi)$ .

Following Gladin et al. [2024], a way to simulate the WIFT-GF. is via an weighted interacting particle system. Let  $\mu_N^{(t)} = \sum_{i=1}^N w_i^{(t)} \delta_{x_i^{(t)}}$  be a weighted empirical measure at iteration  $t$ . Write  $\mathbf{X}_t = (x_1^{(t)}, \dots, x_N^{(t)})$  and  $\mathbf{w}^{(t)} = (w_1^{(t)}, \dots, w_N^{(t)})$ . Denote  $m_\pi = \int k(x, \cdot) d\pi$  as the kernel mean embedding. First evolve the Wasserstein gradient term,

$$x_i^{(t+1)} = x_i^{(t)} - \tau \alpha \left[ \sum_{j=1}^N w_j^{(t)} \nabla_2 k(x_j^{(t)}, x_i^{(t)}) - \nabla m_\pi(x_i^{(t)}) \right]. \quad (1)$$

Then, evolve the IFT gradient term, by solving the following optimal weights: let  $\eta = \beta\tau$ ,

$$\mathbf{w}^{(t+1)} = \arg \min_{\mathbf{w} \in \Delta^N} \left\{ \frac{1}{2} \text{MMD}^2 \left( \sum_{i=1}^N w_i \delta_{x_i^{(t+1)}}, \pi \right) + \frac{1}{2\eta} \text{MMD}^2 \left( \sum_{i=1}^N w_i \delta_{x_i^{(t+1)}}, \mu_N^{(t)} \right) \right\}. \quad (2)$$

Interestingly, letting  $K^{(t+1)} = K(\mathbf{X}_{t+1}) \in \mathbb{R}^{N \times N}$ , the weight problem is

$$\min_{\mathbf{w} \in \Delta^N} \left\{ \frac{1}{2} \mathbf{w}^\top K^{(t+1)} \mathbf{w} - m_\pi(\mathbf{X}_{t+1})^\top \mathbf{w} + \frac{1}{2\eta} (\mathbf{w} - \mathbf{w}^{(t)})^\top K^{(t+1)} (\mathbf{w} - \mathbf{w}^{(t)}) \right\},$$

By definition of kernel quadrature, there is  $m_\pi(\mathbf{X}_{t+1}) = K^{(t+1)} \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})$ . And the solution is, when ignoring the simplex constraint,

$$\mathbf{w}^{(t+1)} = \frac{1}{1 + \eta} \mathbf{w}^{(t)} + \frac{\eta}{1 + \eta} \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}).$$

In contrast, the update scheme for the standard Euclidean gradient is:

$$x_i^{(t+1)} = x_i^{(t)} - \tau \alpha w_i \left[ \sum_{j=1}^N w_j^{(t)} \nabla_2 k(x_j^{(t)}, x_i^{(t)}) - \nabla m_\pi(x_i^{(t)}) \right], \quad \mathbf{w}^{(t+1)} = \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}). \quad (3)$$

Comparing the two update schemes, we note that the two only differs by a weight factor  $w_i$ .

**Definition 2.1** (Weighted Euclidean Geometry). *For a positive weight vector  $\mathbf{w} = (w_1, \dots, w_N) \in \mathbb{R}_{>0}^N$ , we equip the particle space  $(\mathbb{R}^d)^N$  with the  $\mathbf{w}$ -weighted Euclidean inner product  $\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{w}} := \sum_{i=1}^N w_i \langle u_i, v_i \rangle$ , with the induced norm  $\|\mathbf{u}\|_{\mathbf{w}}^2 := \sum_{i=1}^N w_i \|u_i\|_2^2$ . For a differentiable function  $f : (\mathbb{R}^d)^N \rightarrow \mathbb{R}$ , its gradient with respect to the  $\mathbf{w}$ -weighted Euclidean geometry, denoted by  $\text{grad}_{\mathbf{w}} f$ , is defined through*

$$Df(\mathbf{X})[\mathbf{u}] = \langle \text{grad}_{\mathbf{w}} f(\mathbf{X}), \mathbf{u} \rangle_{\mathbf{w}}, \quad \forall \mathbf{u} \in (\mathbb{R}^d)^N.$$

Equivalently, its  $i$ -th component is  $[\text{grad}_{\mathbf{w}} f(\mathbf{X})]_i = \frac{1}{w_i} \nabla_{x_i} f(\mathbf{X})$ .

**Proposition 2.2** (Smoothness under Euclidean and weighted Euclidean geometries). *Let  $\mathbf{w} \in \Delta^N$  with  $w_i > 0$  for every  $i$  and  $\sum_{i=1}^N w_i = 1$ . Suppose that  $k$  is symmetric and twice continuously differentiable, with  $\sup_{x, y \in \mathbb{R}^d} \|\nabla_{11}^2 k(x, y)\|_{\text{op}} = C_1 < \infty$  and  $\sup_{x, y \in \mathbb{R}^d} \|\nabla_{12}^2 k(x, y)\|_{\text{op}} = C_2 < \infty$ . For fixed  $\mathbf{w}$ , define*

$$E(\mathbf{X}, \mathbf{w}) = \frac{1}{2} \text{MMD}^2 \left( \sum_{i=1}^N w_i \delta_{x_i}, \pi \right).$$

*Then  $E(\cdot, \mathbf{w})$  is  $\max_{1 \leq i \leq N} w_i (2C_1 + C_2)$ -smooth under the standard Euclidean geometry; and is  $(2C_1 + C_2)$ -smooth under the  $\mathbf{w}$ -weighted Euclidean geometry.*

## 2.1 Convergence

**Theorem 2.3** (Monotone improvement). *At iteration  $t \in \mathbb{N}$ , let  $\mu_N^{(t)} = \sum_{i=1}^N w_i^{(t)} \delta_{x_i^{(t)}}$  with  $\mathbf{X}_t = (x_1^{(t)}, \dots, x_N^{(t)})$ , and  $\mathbf{w}^{(t)} = (w_1^{(t)}, \dots, w_N^{(t)})$ .*

*Assume that  $k$  is twice continuously differentiable, that  $K(\mathbf{X}_t)$  is positive definite for every  $t$ , and that the weights remain uniformly positive, namely  $w_i^{(t)} \geq \underline{w} > 0$  for every  $i, t$ . Assume moreover that, for every admissible  $\mathbf{w}$ , the map  $\mathbf{X} \mapsto \mathcal{E}(\mathbf{X}, \mathbf{w})$  has an  $L_X$ -Lipschitz gradient on a set containing all iterates. If  $\tau\alpha \leq 2\underline{w}/L_X$ , then for every  $t \geq 0$ ,*

$$\text{MMD} \left( \mu_N^{(t+1)}, \pi \right) \leq \text{MMD} \left( \mu_N^{(t)}, \pi \right). \quad (4)$$

**Lemma 2.4** (Uniform control of the kernel quadrature residual). *Suppose that the assumptions of Theorem 2.3 hold and that the initial weights are chosen as  $\mathbf{w}^{(0)} = \mathbf{w}_{\text{KQ}}(\mathbf{X}_0)$ . Then, for every  $t \geq 0$ ,  $\text{MMD} \left( \sum_{i=1}^N w_{\text{KQ},i}(\mathbf{X}_t) \delta_{x_i^{(t)}}, \pi \right) \leq \text{MMD} \left( \sum_{i=1}^N w_{\text{KQ},i}(\mathbf{X}_0) \delta_{x_i^{(0)}}, \pi \right)$ .*

**Theorem 2.5** (Convergence to the kernel quadrature error). *Suppose that the assumptions of Theorem 2.3 hold, and that the initial weights are chosen as  $\mathbf{w}^{(0)} = \mathbf{w}_{\text{KQ}}(\mathbf{X}_0)$ . Then, the WIFT iterates satisfy*

$$\text{MMD}^2(\mu_N^{(t)}, \pi) \leq \frac{1}{(1+\eta)^{2t}} \text{MMD}^2(\mu_N^{(0)}, \pi) + \left(1 - \frac{1}{(1+\eta)^{2t}}\right) \text{MMD}^2 \left( \sum_{i=1}^N w_{\text{KQ},i}(\mathbf{X}_0) \delta_{x_i^{(0)}}, \pi \right).$$

**Remark 2.6.** *Theorem 2.5 separates the optimization error from the statistical error of the initial kernel quadrature rule. In particular, if  $\text{MMD}(\sum_{i=1}^N w_{\text{KQ},i}(\mathbf{X}_0) \delta_{x_i^{(0)}}, \pi) = O_{\mathbb{P}}(N^{-s/d})$ , then*

$$\text{MMD}(\mu_N^{(t)}, \pi) \leq \frac{1}{(1+\eta)^t} \text{MMD}(\mu_N^{(0)}, \pi) + O_{\mathbb{P}}(N^{-s/d}).$$

*Thus, the optimization error decays geometrically until it reaches the kernel quadrature error floor. If, in addition,  $\text{MMD}(\mu_N^{(0)}, \pi) = O_{\mathbb{P}}(1)$ , then choosing  $t \geq \frac{s}{d \log(1+\eta)} \log N$  is sufficient to obtain  $\text{MMD}(\mu_N^{(t)}, \pi) = O_{\mathbb{P}}(N^{-s/d})$ .*

## 3 Proofs

*Proof of Proposition 2.2.* Recall that

$$E(\mathbf{X}, \mathbf{w}) = \frac{1}{2} \sum_{i,j=1}^N w_i w_j k(x_i, x_j) - \sum_{i=1}^N w_i m_{\pi}(x_i) + \text{const.}$$

For any  $\mathbf{u} = (u_1, \dots, u_N) \in (\mathbb{R}^d)^N$ , direct differentiation gives

$$\begin{aligned} \nabla_{\mathbf{X}}^2 E(\mathbf{X}, \mathbf{w})[\mathbf{u}, \mathbf{u}] &= \sum_{i,j=1}^N w_i w_j u_i^{\top} \nabla_{11}^2 k(x_i, x_j) u_i \\ &\quad + \sum_{i,j=1}^N w_i w_j u_i^{\top} \nabla_{12}^2 k(x_i, x_j) u_j - \sum_{i=1}^N w_i u_i^{\top} \nabla^2 m_{\pi}(x_i) u_i. \end{aligned}$$

Since  $m_\pi(x) = \int k(x, y) d\pi(y)$  and  $\pi$  is a probability measure, we have

$$\|\nabla^2 m_\pi(x)\|_{\text{op}} = \left\| \int \nabla_{11}^2 k(x, y) d\pi(y) \right\|_{\text{op}} \leq \sup_{x, y \in \mathbb{R}^d} \|\nabla_{11}^2 k(x, y)\|_{\text{op}} = C_1.$$

Using  $\sum_{j=1}^N w_j = 1$ , it follows that

$$\left| \sum_{i,j=1}^N w_i w_j u_i^\top \nabla_{11}^2 k(x_i, x_j) u_i - \sum_{i=1}^N w_i u_i^\top \nabla^2 m_\pi(x_i) u_i \right| \leq 2C_1 \sum_{i=1}^N w_i \|u_i\|^2.$$

For the mixed term,

$$\left| \sum_{i,j=1}^N w_i w_j u_i^\top \nabla_{12}^2 k(x_i, x_j) u_j \right| \leq \sup_{x, y \in \mathbb{R}^d} \|\nabla_{12}^2 k(x, y)\|_{\text{op}} \left( \sum_{i=1}^N w_i \|u_i\| \right)^2 = C_2 \left( \sum_{i=1}^N w_i \|u_i\| \right)^2.$$

By Cauchy–Schwarz and  $\sum_{i=1}^N w_i = 1$ , we have  $\left( \sum_{i=1}^N w_i \|u_i\| \right)^2 \leq \sum_{i=1}^N w_i \|u_i\|^2$ . Therefore,

$$|\nabla_{\mathbf{X}}^2 E(\mathbf{X}, \mathbf{w})[\mathbf{u}, \mathbf{u}]| \leq (2C_1 + C_2) \sum_{i=1}^N w_i \|u_i\|^2 = (2C_1 + C_2) \|\mathbf{u}\|_{\mathbf{w}}^2.$$

The last step holds by the definition of  $\mathbf{w}$ -weighted Euclidean geometry in Definition 2.1. This proves smoothness under the weighted Euclidean geometry.

For the standard Euclidean geometry, since  $\sum_{i=1}^N w_i \|u_i\|^2 \leq (\max_{1 \leq i \leq N} w_i) \sum_{i=1}^N \|u_i\|^2 = (\max_{1 \leq i \leq N} w_i) \|\mathbf{u}\|_2^2$ . The proof is concluded.  $\square$

*Proof of Theorem 2.3.* For fixed  $\mathbf{X}$  and  $\mathbf{w}$ , let  $g_i(\mathbf{X}, \mathbf{w}) = \sum_{j=1}^N w_j \nabla_2 k(x_j, x_i) - \nabla m_\pi(x_i)$ . Define the MMD energy by

$$\mathcal{E}(\mathbf{X}, \mathbf{w}) = \frac{1}{2} \text{MMD}^2 \left( \sum_{i=1}^N w_i \delta_{x_i}, \pi \right), \quad (5)$$

Differentiating the MMD energy with respect to  $x_i$  and using symmetry of the kernel gives  $\nabla_{x_i} \mathcal{E}(\mathbf{X}, \mathbf{w}) = w_i g_i(\mathbf{X}, \mathbf{w})$ . Hence, the position update can be written as  $x_i^{(t+1)} = x_i^{(t)} - \frac{\tau\alpha}{w_i^{(t)}} \nabla_{x_i} \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)})$ . Thus, the Wasserstein substep is a diagonally preconditioned gradient step for  $\mathbf{X} \mapsto \mathcal{E}(\mathbf{X}, \mathbf{w}^{(t)})$ . By the  $L_X$ -smoothness of  $\mathcal{E}(\cdot, \mathbf{w}^{(t)})$ , the descent lemma gives

$$\begin{aligned} \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t)}) &\leq \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)}) + \sum_{i=1}^N \left\langle \nabla_{x_i} \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)}), x_i^{(t+1)} - x_i^{(t)} \right\rangle + \frac{L_X}{2} \sum_{i=1}^N \|x_i^{(t+1)} - x_i^{(t)}\|^2 \\ &= \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)}) - \tau\alpha \sum_{i=1}^N \left( w_i^{(t)} - \frac{L_X \tau\alpha}{2} \right) \|g_i(\mathbf{X}_t, \mathbf{w}^{(t)})\|^2. \end{aligned} \quad (6)$$

Since  $w_i^{(t)} \geq \underline{w}$  and  $\tau\alpha \leq 2\underline{w}/L_X$ , we have  $w_i^{(t)} - L_X \tau\alpha/2 \geq 0$ . Therefore,  $\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t)}) \leq \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)})$ . This proves that the particle update in Eq. (1) is decreasing the MMD value.

We next consider the weight update. For fixed  $\mathbf{X}$ , the dependence of the MMD energy on  $\mathbf{w}$  is quadratic:  $\mathcal{E}(\mathbf{X}, \mathbf{w}) = \frac{1}{2} \mathbf{w}^\top K(\mathbf{X}) \mathbf{w} - m_\pi(\mathbf{X})^\top \mathbf{w} + \text{const.}$  Since  $K(\mathbf{X})$  is positive definite, the unique minimizer is  $\mathbf{w}_{\text{KQ}}(\mathbf{X}) = K(\mathbf{X})^{-1} m_\pi(\mathbf{X})$ . Completing the square gives

$$\mathcal{E}(\mathbf{X}, \mathbf{w}) = \mathcal{E}(\mathbf{X}, \mathbf{w}_{\text{KQ}}(\mathbf{X})) + \frac{1}{2} (\mathbf{w} - \mathbf{w}_{\text{KQ}}(\mathbf{X}))^\top K(\mathbf{X}) (\mathbf{w} - \mathbf{w}_{\text{KQ}}(\mathbf{X})). \quad (7)$$

By the weight update,

$$\mathbf{w}^{(t+1)} - \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}) = \frac{1}{1+\eta} (\mathbf{w}^{(t)} - \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})). \quad (8)$$

On the other hand, for fixed particle locations  $\mathbf{X}_{t+1}$ , the KQ decomposition gives

$$\begin{aligned} \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t+1)}) - \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})) \\ = \frac{1}{2} (\mathbf{w}^{(t+1)} - \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}))^\top K^{(t+1)} (\mathbf{w}^{(t+1)} - \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})). \end{aligned} \quad (9)$$

Substituting (8) into (9) yields

$$\begin{aligned} \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t+1)}) - \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})) \\ = \frac{1}{2(1+\eta)^2} (\mathbf{w}^{(t)} - \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}))^\top K^{(t+1)} (\mathbf{w}^{(t)} - \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})) \\ = \frac{1}{(1+\eta)^2} [\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t)}) - \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}))]. \end{aligned} \quad (10)$$

Since  $(1+\eta)^{-2} \leq 1$  and  $\mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})$  minimizes  $\mathbf{w} \mapsto \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w})$ , it follows that  $\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t+1)}) \leq \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t)})$ . Combining the two substeps yields

$$\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t+1)}) \leq \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t)}) \leq \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)}). \quad (11)$$

Hence,  $\mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)})$  is nonincreasing, and the proof is concluded.  $\square$

*Proof of Lemma 2.4.* Recall the definition that  $\mathcal{E}(\mathbf{X}, \mathbf{w}) = \frac{1}{2} \text{MMD}^2 \left( \sum_{i=1}^N w_i \delta_{x_i}, \pi \right)$ . For any fixed particle locations  $\mathbf{X}_t$ , the kernel quadrature weights  $\mathbf{w}_{\text{KQ}}(\mathbf{X}_t)$  minimize the MMD energy over the weights. Hence,  $\mathcal{E}(\mathbf{X}_t, \mathbf{w}_{\text{KQ}}(\mathbf{X}_t)) \leq \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)})$ . On the other hand, by the monotonicity of the WIFT iteration established in Theorem 2.3, we have  $\mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)}) \leq \mathcal{E}(\mathbf{X}_0, \mathbf{w}^{(0)})$ . Since  $\mathbf{w}^{(0)} = \mathbf{w}_{\text{KQ}}(\mathbf{X}_0)$ , we have  $\mathcal{E}(\mathbf{X}_0, \mathbf{w}^{(0)}) = \mathcal{E}(\mathbf{X}_0, \mathbf{w}_{\text{KQ}}(\mathbf{X}_0))$ . Combining the previous inequalities gives

$$\mathcal{E}(\mathbf{X}_t, \mathbf{w}_{\text{KQ}}(\mathbf{X}_t)) \leq \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)}) \leq \mathcal{E}(\mathbf{X}_0, \mathbf{w}^{(0)}) = \mathcal{E}(\mathbf{X}_0, \mathbf{w}_{\text{KQ}}(\mathbf{X}_0)). \quad (12)$$

The proof is thus concluded.  $\square$

*Proof of Theorem 2.5.* From the exact contraction of the weight update in Eq. (10), together with descent of the Wasserstein position update, we have

$$\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t+1)}) \leq \frac{1}{(1+\eta)^2} \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)}) + \left(1 - \frac{1}{(1+\eta)^2}\right) \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})). \quad (13)$$

By Lemma 2.4,  $\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})) \leq \mathcal{E}(\mathbf{X}_0, \mathbf{w}_{\text{KQ}}(\mathbf{X}_0))$ , and therefore

$$\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}^{(t+1)}) \leq \frac{1}{(1+\eta)^2} \mathcal{E}(\mathbf{X}_t, \mathbf{w}^{(t)}) + \left(1 - \frac{1}{(1+\eta)^2}\right) \mathcal{E}(\mathbf{X}_0, \mathbf{w}_{\text{KQ}}(\mathbf{X}_0)). \quad (14)$$

The proof is concluded.  $\square$

## References

- Egor Gladin, Pavel Dvurechensky, Alexander Mielke, and Jia-Jie Zhu. Interaction-force transport gradient flows. *Advances in Neural Information Processing Systems*, 37:14484–14508, 2024.
- Jia-Jie Zhu. Inclusive kl minimization: A wasserstein-fisher-rao gradient flow perspective. *arXiv preprint arXiv:2411.00214*, 2024.
- Jia-Jie Zhu and Alexander Mielke. Kernel approximation of fisher-rao gradient flows. *arXiv preprint arXiv:2410.20622*, 2024.